higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
NQF LEVEL 3

(12041023)

4 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.
5. Answer each question on a new page

NCV

QUESTION 1: FUNDAMENTALS OF ELECTRONICS

1.1 Technicians rely on measuring instruments for their accuracy.

Name FIVE ways that you would use to take care of a digital multimeter. (5)

1.2 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–E) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.2.1	Free electron	A	P-type semiconductor
1.2.2	Acceptor doping	B	wide energy gap
1.2.3	Covalent bonding	C	electrically charged atom
1.2.4	Insulator	D	unattached electron
1.2.5	Ion	E	mutual sharing of electrons
		F	extrinsic semiconductor
		G	impurity

(5 × 1) (5)

1.3 1.3.1 Explain the basic operation of a photo-diode. (2)

1.3.2 Semiconductor diodes operation can be displayed as a characteristic curve.

Draw the characteristic curve of a Zener diode and label it. (5)

1.4 1.4.1 A transistor can be configured as an amplifier to suit different characteristics.

Name TWO characteristics of a common emitter amplifier circuit. (2)

1.4.2 Calculate the DC current gain of a common emitter amplifier circuit, where the I_B = 0,04 mA and the I_C = 2 mA (3)

1.4.3 When choosing a transistor to be used as an amplifier, the symbol $\mathbf{h_{FE}}$ is used in a technical manual.

What characteristic does this symbol indicate? (1)

1.5 1.5.1 Power supplies convert AC to DC.

Explain the operation of a half-wave rectifier circuit supplying a load R_L. (3)

1.5.2 The half wave rectifier circuits output needs to be filtered to form smooth DC.

Explain how an inductor and capacitor can be configured into the circuit as filter components. (2)

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1.6 1.6.1 Draw the labelled circuit symbol of an N-channel J-FET. (3)

1.6.2 MOSFETs are sensitive, modified semiconductor transistors.

Explain how these components make use of a metal oxide layer in their construction. (2)

1.7 Refer to FIGURE 1 below of an operational amplifier and answer the questions.

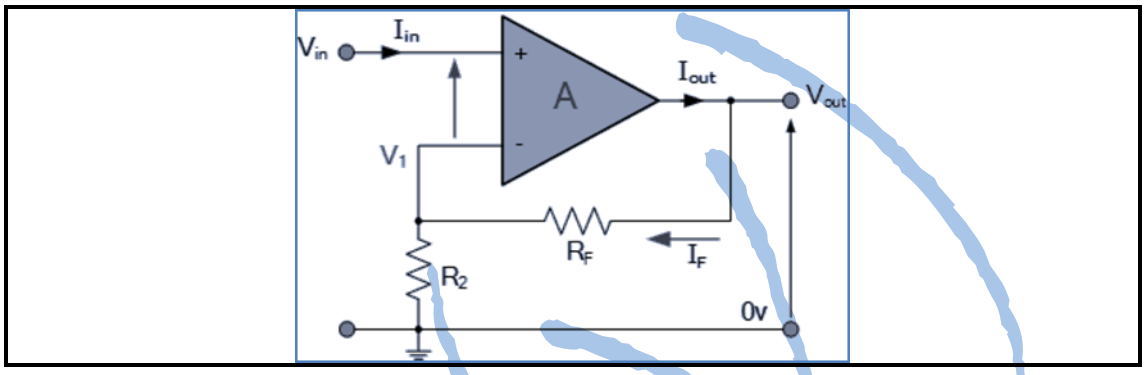


FIGURE 1

1.7.1 Identify the Op-amp circuit. (1)

1.7.2 What is the function of resistor R_f? (1)

1.7.3 Name TWO characteristics of this circuit. (2)

1.7.4 Calculate the output voltage of the circuit if the input voltage is 0,5 V and the resistors values are, Rin = 2 kΩ and Rf = 20 kΩ (3)

[40]

QUESTION 2: DIGITAL ELECTRONICS

2.1 Convert between the following number systems:

2.1.1 59_{10} to hexadecimal

2.1.2 135_8 to decimal

2.1.3 11101110_2 to octal

(3 × 3) (9)

2.2 Multiply the following binary numbers:

$10101_2 \times 101_2$ (3)

2.3 Divide the following binary numbers:

$10111_2 \div 110_2$ (3)

2.4 Subtract the following binary numbers:

$10100_2 - 101_2$ (3)

2.5 Digital counters are a sequenced register of flip-flops that have the ability to count numbers.

2.5.1 Explain the basic operation of a binary asynchronous decade counter. (3)

2.5.2 Draw a labelled block diagram of a 3-bit asynchronous down counter. (4)
[25]

QUESTION 3: ASSEMBLING A PERSONAL COMPUTER

3.1 Identify four external components of a typical desktop computer that will ensure its operation. (4)

3.2 The motherboard is a web of connectivity to all internal components.

Name THREE types of sockets found on the motherboard inside the tower. (3)

3.3 A mouse of a computer is not operating well.

Name THREE common problems that could affect its operation. (3)

3.4 A computer has come in for repair and you find that the motherboard has electronic faults on it.

Discuss how you would remove and replace the motherboard on the computer. (5)
[15]

QUESTION 4: TRANSDUCERS

4.1 As an instrumentation technician, you have to design a control process using transducers.

Name FIVE requirements or factors you would consider when choosing the appropriate transducer. (5)

4.2 Inductive transducers can be modified to measure velocity or rotational speed.

Explain how this is accomplished, referring to a basic drawn sketch. (5)

[10]

QUESTION 5: LADDER LOGIC – PROGRAMMABLE LOGIC CONTROLLERS

5.1 Older relay control systems were replaced by PLCs.

Name FOUR reasons why you would prefer using a PLC control system instead of relays. (4)

5.2 Design a ladder logic diagram that will operate like an XOR logic gate. (6)

[10]

TOTAL: 100

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